

INTERCHANGING INTEGRATION AND DIFFERENTIATION

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This is very standard. See, for example, [Fol99, p 56].

In the following we have a function $f(x, t)$ of two variables $x \in E \subset \mathbb{R}^d$ and $t \in [a, b]$. You may want to think of the case $d = 1$. Note that all integrals are with respect to the x variable.

Let

$$F(t) = \int_E f(x, t) dx.$$

In particular we assume that $f(x, t)$ is integrable in x for each $t \in [a, b]$. We want to know if:

- (1) Assuming $f(x, t)$ is continuous in t for each $x \in E$, does it follow that $F(t)$ is continuous?
- (2) Assuming $f(x, t)$ is differentiable in t for each $x \in E$ does it follow that $F'(t)$ exists and $F'(t) = \int_E \frac{\partial}{\partial t} f(x, t) dx$?

The answer is “yes”, provided that the functions $f(\cdot, t)$, respectively $\frac{\partial}{\partial t} f(\cdot, t)$, are uniformly integrable over E . That is, provided they are bounded by some positive integrable function independent of t .

The proof is straightforward using

- (1) the dominated convergence theorem, and
- (2) the observation that the relevant limit exists at t_0 if it exists for any sequence $t_n \rightarrow t_0$ and is independent of the sequence.

Theorem. Suppose $E \subset \mathbb{R}^d$, $f : E \times [a, b] \rightarrow \mathbb{R}$, and $f(x, t)$ is integrable in x for each $t \in [a, b]$. Let

$$F(t) = \int_E f(x, t) dx.$$

Suppose $f(x, t)$ is continuous in t for each $x \in E$. Suppose for all $x \in E$ and $t \in [a, b]$ that $|f(x, t)| \leq g(x)$ where $g : E \rightarrow \mathbb{R}$ is integrable. Then $F(t)$ is continuous in t , i.e.

$$(1) \quad \lim_{t \rightarrow t_0} \int_E f(x, t) dx = \int_E f(x, t_0) dx \quad \forall t_0 \in [a, b].$$

Suppose $\frac{\partial}{\partial t} f(x, t)$ exists for all t and all x . Suppose for all $x \in E$ and $t \in [a, b]$ that $|\frac{\partial}{\partial t} f(x, t)| \leq k(x)$ where $k : E \rightarrow \mathbb{R}$ is integrable. Then F is differentiable and

$$(2) \quad \frac{\partial}{\partial t} \bigg|_{t=t_0} \int_E f(x, t) dx = \int_E \frac{\partial}{\partial t} \bigg|_{t=t_0} f(x, t) dx \quad \forall t_0 \in [a, b].$$

Proof. Let $(t_n)_{n \geq 1}$ be any sequence in $[a, b]$ such that $t_n \rightarrow t_0$. Let $f_n(x) = f(x, t_n)$ and $f(x) = f(x, t_0)$. Since $f_n(x) \rightarrow f(x)$ and $|f_n(x)| \leq g(x)$ for all x , where $\int_E g < \infty$, it follows from the dominated convergence theorem that

$$\lim_{n \rightarrow \infty} \int_E f(x, t_n) dx = \int_E f(x, t_0) dx \quad \forall t_0 \in [a, b].$$

Since the right side is independent of the sequence (t_n) , (1) follows.

Again let $(t_n)_{n \geq 1}$ be any sequence in $[a, b]$ such that $t_n \rightarrow t_0$. Then

$$h_n(x) := \frac{f(x, t_n) - f(x, t_0)}{t_n - t_0} \rightarrow \frac{\partial f}{\partial t}(x, t_0) \quad \text{as } t_n \rightarrow t_0.$$

By the mean value theorem, for some ξ_n between t_n and t_0 ,

$$|h_n(x)| = \left| \frac{\partial f}{\partial t}(x, \xi_n) \right| \leq k(x).$$

It follows by the dominated convergence theorem that

$$\begin{aligned} \int h_n(x) \, dx &\rightarrow \int_E \frac{\partial f}{\partial t}(x, t_0) \, dx \quad \text{as } t_n \rightarrow t_0, \\ \text{i.e. } \frac{\int_E f(x, t_n) - \int_E f(x, t_0)}{t_n - t_0} &\rightarrow \int_E \frac{\partial f}{\partial t}(x, t_0) \, dx \quad \text{as } t_n \rightarrow t_0. \end{aligned}$$

Since this is true for any sequence $(t_n)_{n \geq 1}$ from $[a, b]$, (2) follows. $\square$

REFERENCES

[Fol99] Gerald B. Folland, *Real analysis*, 2nd ed., 1999.